

SUMMARY:

Machine Learning Engineer with 3+ years building and operating production ML and GenAI systems. Skills include time series forecasting, NLP and LLMs, RAG and agents, and real-time anomaly and fraud detection. MLOps with Databricks, Spark, feature stores, MLflow registry, CI/CD, monitoring for drift and latency, AWS, Azure, Docker, Kubernetes. Proficient in Python and SQL. Delivers secure and explainable models with guardrails and SHAP, and business dashboards in Tableau, Power BI, and Streamlit. Certified Azure Data Scientist.

SKILLS:

- Methodologies:** SDLC, Agile, Waterfall
- Programming Languages:** Python, R, Java, SQL
- Machine Learning & Deep Learning:** XGBoost, LightGBM, Random Forest, Logistic Regression, Linear Regression, Clustering, Time Series (ARIMA, Prophet), A/B Testing, Feature Engineering, Fraud Detection, LSTM, CNN, Autoencoders, GANs, Transformers
- ML Frameworks & Libraries:** scikit-learn, TensorFlow, PyTorch, Keras
- Data Science Tools:** Pandas, NumPy, Matplotlib, Seaborn
- Big Data Technologies:** Hadoop, Spark, PySpark, Apache Kafka, Airflow
- Cloud Platforms:** AWS (S3, EC2, SageMaker), Azure (Synapse, Data Factory, Azure ML), GCP, Microsoft Fabric, Databricks, Snowflake
- DevOps & MLOps Tools:** Docker, Kubernetes, Git, Jenkins, MLflow, CI/CD
- NLP & Generative AI:** Hugging Face Transformers, LangChain, Prompt Engineering, LLM Fine-Tuning, GPT, BERT, RAG Architectures, LangGraph, vLLM, Vector Databases (FAISS, Pinecone), Rerankers (Cohere, ColBERT), Retrieval-Augmented Generation (RAG)
- Computer Vision & Visualization:** OpenCV, Tableau, Power BI

EDUCATION:

Masters in Computer Science and Data Science	May 2025
North Carolina State University, Raleigh, NC	
Bachelor of Technology in Electronics and Communication Engineering	May 2021
GITAM University, Hyderabad, India	

EXPERIENCE:

Cardinal Health, NC | Nov 2024 – Current | ML Engineer

- Built a demand sensing and replenishment service using quantile forecasting (P50 and P90) at SKU by location with a short horizon optimizer which reduced stockouts by 20-25% and cut overstock by 12-15% in initial channels.
- Delivered a recall and safety workflow with a deterministic matcher (NDC, UDI, lot) and a RAG-based extractor for unstructured notices with JSON schema validation and citations; reduced triage time by approximately 35% and manual review by approximately 40% in early rollouts by achieving more than 98% exact-match ID.
- Engineered batch and streaming pipelines and a feature store with Airflow, AWS Glue, and Apache Spark (Databricks) to unify EHR, inventory, and distributor feeds; added domain features (seasonality, regional disease trends, formulary changes) which reduced data latency by 40% and improved forecast accuracy, lowering MAPE from 18% to 9%.
- Standardized MLOps with MLflow Model Registry, AWS CloudWatch drift alerts, and canary flags and enabled auto retraining when accuracy dropped below SLAs, sustained 99% uptime.

Tata Consultancy Services, India | June 2022 – June 2023 | Data Scientist

- Built a customer support RAG assistant using hybrid search (BM25 and dense retrieval), LangChain, prompt engineering, and LoRA fine-tuned LLMs on 200,000 feedback records for KYC, disputes, and chargebacks FAQs, improved NPS by 12% and reduced resolution time by 15% in A-B tests, implemented PII redaction and prompt safety guardrails.
- Replaced autoencoders with graph and sequence aware fraud modeling by computing GraphSAGE embeddings on a transaction entity graph (card, device, merchant) and pairing them with an XGBoost classifier, plus a Temporal Convolutional Network for short behavior sequences (velocity, MCC, CNP signals), increased precision at 95% recall by 4.5% and reduced scoring latency by 20% on Azure Event Hubs and Azure Kubernetes Service.
- Established online evaluation and controlled rollout for ML services with shadow deployments, canary releases, and feature flags, with automated threshold calibration to hold false positive rate below 1%, reduced production lead time by 40% and decreased rollbacks.
- Built PySpark feature pipelines and a reusable Feature Store on Azure Databricks with Delta Lake and MLflow Model Registry for fraud, churn, loan default, and RAG use cases, standardized Azure DevOps CI and CD and added drift alerts, speeding analysis by 15%, cutting model refresh cycles by 30%, and maintaining 99% uptime.

Tata Consultancy Services, India | June 2021 – May 2022 | Junior Data Scientist

- Assisted in building churn and loan default models by engineering seasonality, trend, and holiday features in Snowflake using SQL and Prophet calendars, trained LightGBM for churn and CatBoost for loan default, produced SHAP explanations in Power BI for stakeholders, resulting in a 4% to 6% reduction in churn and a 3% to 4% decrease in high-risk approvals.
- Built a two-stage next best offer system using two tower neural retrieval for candidate generation with approximate nearest neighbor search and DeepFM to re rank top K using eligibility and recent behavior features, served real time scores via GraphQL, A/B tests delivered a 7% to 9% lift in approved applications.
- Streamlined SQL ETL with incremental change data capture loads and data quality checks, reducing runtime by 20% and raising on time loads to 99% for customer transaction marts powering Power BI dashboards.

PROJECTS:

Agentic Eligibility and Compliance Copilot

- Implemented OCR to LLM JSON extraction with schema validation and retrieval augmented generation over domain policies and standard operating procedures; used LangChain and LangGraph tools for identity checks, calculations, and checklists, and RoBERTa for routing; achieved 94% accuracy and 98% exact match on key fields; built a Streamlit review UI.

NLP to CLI Copilot

- Translates natural language requests into safe shell commands using transformer models (GPT, CodeBERT, T5) with context awareness of environment and user preferences and multi step chaining; achieved 85% task accuracy and a 15% relevance improvement; served via FastAPI for concurrent requests.

Adaptive Anomaly Detection

- Built a streaming anomaly detector on more than 100,000 synthetic and realistic events using Anomaly Transformer with a gradient boosting head; achieved AUC 0.91 with calibrated thresholds; containerized with Docker and exposed a FastAPI service with a Kafka simulator; ran 30% faster than the baseline.

CERTIFICATES:

- DP-100: Microsoft Certified: Azure Data Scientist Associate**
- DP-900: Microsoft Certified: Azure Data Fundamentals**
- AI-900: Microsoft Certified: Azure AI Fundamentals**
- AZ-900: Microsoft Certified: Azure Fundamentals**